

Phase 1: Data Engineering & Quality Control (Summary)

In this phase, I worked on improving the database structure and checking the quality of the data. The goal was to turn raw data into a reliable source of truth by finding and fixing errors that could have affected the accuracy of the results.

1. Import Optimization & Schema Safety

Redundant Data Removal: During the import process, I removed columns that were not needed for the analysis. This helped reduce the size of the database and made queries run faster.

Data Type Precision: I changed the `customer_zip_code_prefix` column to `VARCHAR(20)`. This step was important to keep leading zeros in postal codes and avoid formatting problems such as unwanted thousand separators.

Storage Efficiency: I adjusted `VARCHAR` lengths manually to make sure the data was stored correctly without using more space than necessary.

2. Text and Financial Standardization

Geographical Normalization: In the `customers` and `sellers` tables, I changed all city names to uppercase using the `UPPER` function. This made sure that city names were grouped correctly, removing problems caused by inconsistent letter cases.

Currency Accuracy: In the SQL Views, I applied a two-decimal format (`DECIMAL(10,2)`). This prevents small rounding errors that are common with floating-point numbers and keeps all revenue totals accurate.

3. Advanced Duplicate Resolution

Problem Identification: In the `order_reviews` table, I found duplicate rows where the same `review_id` appeared more than once but with different timestamps.

The Solution: I used the `ROW_NUMBER()` window function, grouped by `review_id` and sorted by date from newest to oldest. This kept only the most recent record for each review and removed all older duplicates, making sure the customer satisfaction data was accurate.

4. SQL Views — The Analytical Layer

I created a set of SQL Views to keep the raw data separate from the analytical layer. These views prepare the data automatically before it is used in visualizations:

v_payments_pretty: Cleans up payment method names by replacing underscores with spaces and applies a standard currency format.

v_category_translation: Replaces technical Portuguese category names with readable English translations.

v_products_pretty: Prepares a clean product list that makes it easier to group items and run category analysis.

5. NULL Audit & Data Pruning

Audit Results: I checked all tables for missing values. In the products table, I found 610 records with no category name and two records with zero physical dimensions.

Data Cleaning: I used the COALESCE(NULLIF()) technique to handle empty strings that were stored instead of proper NULL values. Records that could not be fixed were excluded to prevent unknown categories from affecting the business results.

6. Future Process Optimization

Strategy: All data transformations were done in SQL to keep full control over the logic. For future larger projects, I plan to use Power Query (M language) as well.

Benefits: Tools like the Column Quality view in Power Query allow faster error detection and more automated data preparation during the visualization phase.